Muhammad Muizz Zafar

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Software Engineer | YAML Configurations | Data Pipelines | Cloud Infrastructure

Skills

Programming Languages:	Java, Python, TypeScript, GoLang, SQL, JavaScript, Kotlin, C#, C++, HTML/CSS, PHP
Tools/Frameworks:	AWS, AWS CDK, CloudFormation, Docker, Splunk, GraphQL, Git (version control), Linux
Structured Data:	YAML, JSON, XML, SQL Schema Validation

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work experience

Devops and Data Validation Developer — May 2024 - Present
Elections Canada — Ottawa, ON

- Built automated Python ETL validation pipelines and configuration-driven CI/CD workflows **(YAML-based)** in Azure DevOps to verify production data movement across **20M+ records**.
- Maintained configuration-driven workflows using **YAML** and version-controlled CI/CD pipelines to ensure consistent data validation across environments.
- Used **Apache Spark** to profile, transform, and validate structured datasets (JSON, SQL tables) of up to **5+ GB**, identifying schema inconsistencies and edge cases in pipeline data.
- Built internal **data analytics dashboards** using **Python, Streamlit, and Plotly Dash** to surface pipeline health, operational metrics, validation trends, and data quality issues for directorate and senior management.
- Coordinated version-controlled database deployments and release processes, improving traceability and reliability of production data infrastructure across shared environments.
- Collaborated in an Agile cross-functional team with cloud engineers to improve **distributed data workflows**, release quality, and production reliability.

Full Stack Developer — September 2023 - April 2024
University of Ottawa — Ottawa, ON

- Developed backend and frontend features for the University emergency alert system using **Laravel**, **Drupal, PHP**, and JavaScript.
- Built and maintained **JAVA** REST APIs and optimized database queries to improve response times for a platform serving **20,000+ users**.
- Implemented automated **unit and end-to-end tests** using Cypress, Jasmine, and JUnit to ensure application reliability.
- Worked closely with cross-functional teams to debug production issues and deliver stable releases.
- Investigated and resolved production issues by debugging backend APIs and database queries, improving system reliability.

Cloud Developer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Developed serverless monitoring workflows using **AWS Lambda, S3, and CloudWatch** to analyze **50,000+ artifacts per day** and support automated vulnerability scanning pipelines for high-volume cloud artifacts.
- Defined and maintained cloud infrastructure using **Golang**, **AWS CDK**, through **CloudFormation** to provision **50+ AWS resources** and improve repeatability of service deployments.
- Implemented concurrent processing logic in **GoLang** using goroutines to improve throughput and strengthen fault handling for **distributed cloud data processing** workloads.
- Integrated vulnerability scanning tools (**Anchore, Grype**) with AWS services to automate security checks and operational monitoring across **50,000+ cloud artifacts per day**.

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Built a serverless automation pipeline using **AWS Lambda, S3, and Athena** to process incoming cloud data events, query metadata, and trigger downstream updates, reducing manual operational effort.
- Configured **Splunk** and **AWS CloudWatch** monitoring, alarms, dashboards, and alerts to track **15+ services** and improve observability of production cloud infrastructure.
- Developed **Docker-based integration tests** and unit tests to improve reliability of distributed serverless data pipelines.

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Built **Docker** containers and automated event ingestion into **PostgreSQL** using **Python** and **GraphQl**, reducing testing time and improving deployment readiness.